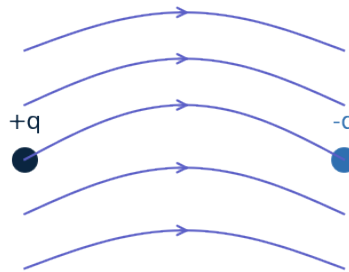


Electrostatics

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Coulomb's Law: force between two point charges $F = k q_1 q_2 / r^2$, where $k = 1/(4 \pi \epsilon_0)$ approx $9 \times 10^9 \text{ N m}^2/\text{C}^2$.
- Electric field $E = F/q_0$, a vector quantity. Field due to a point charge: $E = kq/r^2$, directed radially.
- Electric potential $V = kq/r$ (scalar); potential difference relates to work done per unit charge: $W = q(V_1 - V_2)$. $E = -dV/dr$.
- Gauss's Law: the net electric flux through a closed surface $= q(\text{enclosed})/\epsilon_0$. Useful for finding fields of symmetric charge distributions (sphere, cylinder, infinite sheet).
- Electric dipole: two equal and opposite charges separated by a small distance; dipole moment $p = q \times 2a$, directed from negative to positive charge.
- Capacitance $C = Q/V$. For a parallel plate capacitor, $C = \epsilon_0 A/d$. Capacitors in series: $1/C = 1/C_1 + 1/C_2 + \dots$; in parallel: $C = C_1 + C_2 + \dots$
- Energy stored in a capacitor: $U = (1/2) C V^2 = (1/2) Q V = Q^2/(2C)$.



Electric field lines between a positive and a negative point charge (a dipole).

2. Formula Sheet

Coulomb's law	$F = k q_1 q_2 / r^2, k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$
Electric field (point charge)	$E = kq/r^2$
Electric potential	$V = kq/r$
Gauss's law	$\text{Flux} = q(\text{enclosed}) / \epsilon_0$
Dipole moment	$p = q (2a)$
Field on dipole axis (far away)	$E = 2kp/r^3$
Parallel plate capacitor	$C = \epsilon_0 A / d$
Energy stored in a capacitor	$U = (1/2) C V^2$

3. Practice Problems (22 questions)

- Find the force between two charges of 2 microC and 3 microC separated by 0.3 m ($k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$).
- Find the electric field at a distance of 0.5 m from a point charge of 4 microC.
- Find the potential at a point 0.2 m from a point charge of 5 microC.
- Two point charges +4 microC and -4 microC are separated by 0.02 m, forming a dipole. Find the dipole moment.
- A parallel plate capacitor has plate area 0.02 m² and separation 1 mm. Find its capacitance ($\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$).
- Three capacitors of 2 microF, 3 microF, and 6 microF are connected in series. Find the equivalent capacitance.
- Three capacitors of 2 microF, 3 microF, and 5 microF are connected in parallel. Find the equivalent capacitance.
- A capacitor of 10 microF is charged to a potential difference of 100 V. Find the energy stored in it.
- State Gauss's law and explain how it simplifies finding the electric field of an infinite charged sheet.
- Find the work done in moving a charge of 2 microC through a potential difference of 50 V.
- Two identical charges of 5 microC each are placed 1 m apart. Find the electrostatic potential energy of the system.
- Explain why electric field lines never cross each other.
- A charge of 1 microC is placed at the origin. Find the electric field at a point 1 m away along the x-axis.
- State the principle of superposition for electric fields due to multiple point charges.
- Find the equivalent capacitance of two capacitors of 4 microF each connected first in series, then in parallel.
- A dipole of dipole moment $2 \times 10^{-6} \text{ C m}$ is placed in a uniform electric field of 500 N/C at 30 degrees to the field. Find the torque on it.

17. Explain why the electric field inside a conductor in electrostatic equilibrium is zero.
18. Find the capacitance of a spherical conductor of radius 0.1 m in air ($C = 4\pi\epsilon_0 R$).
19. Two point charges of equal magnitude q are placed at the two corners of an equilateral triangle of side a . Find the direction of the resultant electric field at the third corner (charges are like signs).
20. State two differences between electric potential and electric potential energy.
21. A charge of 6 microC experiences a force of 0.06 N in an electric field. Find the magnitude of the electric field.
22. Explain why the potential energy of an isolated system of two like charges is positive, while that of two unlike charges is negative.

Answer Key

- | | |
|--|---|
| 1. $F = 0.6 \text{ N}$ | 2. $E = 1.44 \times 10^5 \text{ N/C}$ |
| 3. $V = 2.25 \times 10^5 \text{ V}$ | 4. $p = 8 \times 10^{-8} \text{ C m}$ |
| 5. $C \approx 1.77 \times 10^{-10} \text{ F}$ (177 pF) | 6. $C(\text{eq}) = 1 \text{ microF}$ |
| 7. $C(\text{eq}) = 10 \text{ microF}$ | 8. $U = 0.05 \text{ J}$ |
| 9. Conceptual - symmetry lets you choose a Gaussian surface where E is constant, simplifying the flux integral | 10. $W = 1 \times 10^{-4} \text{ J}$ |
| 11. $U \approx 0.225 \text{ J}$ | 12. Conceptual - each point has a unique field direction; crossing would imply two directions at once |
| 13. $E = 9000 \text{ N/C}$ | 14. Conceptual - net field is the vector sum of fields due to each individual charge |
| 15. Series: 2 microF; Parallel: 8 microF | 16. $\tau = 5 \times 10^{-4} \text{ N m}$ |
| 17. Conceptual - free charges redistribute until the internal field cancels any external field | 18. $C \approx 1.11 \times 10^{-11} \text{ F}$ |
| 19. Conceptual - resultant field bisects the angle, directed away from the triangle (for like charges) | 20. Conceptual - potential (V) is work per unit charge (scalar, J/C); potential energy (U) is total work for the given charge (J) |
| 21. $E = 10,000 \text{ N/C}$ | 22. Conceptual - like charges require positive work to bring together (repulsive); unlike charges release energy (attractive), giving negative PE |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.